

Section C: Advanced RAG

C1. You're building a RAG system for a company's product manuals. These manuals have text instructions along with diagrams and photos showing how to assemble the product. A user asks "how do I attach the side panel," and the answer is shown entirely in a diagram, with no accompanying text.

Explain how you would approach this with multimodal RAG. What would go wrong if you used a text-only RAG pipeline here?

For this case, which contains text, diagrams, and photos, I would use a multimodal RAG so that the system can understand the context of the text as well as the image information. This way, the system can easily retrieve the relevant information from the data.

Approach:

1. Separate the text from the diagrams and images.
2. **Chunking:** Split the text into small sections, and keep each diagram tied to its page and section.
3. Save everything in a vector database.
4. When the user types "How do I attach the side panel?", the system converts the question into an embedding.
5. Compare the question embedding with the stored embeddings and return the most relevant text.
6. Retrieve the side-panel diagram even if it has no words, because the diagram has its own embedding.

What goes wrong with text-only RAG?

If we use text-only RAG, it only searches the text embeddings, but the answer is entirely contained in the diagram, so the diagram gets ignored.

As a result:

- The relevant diagram is not retrieved.
- The system may retrieve other unrelated text from nearby data.

C2. Briefly compare the two approaches to multimodal retrieval covered in lecture: unified embedding space vs separate indexes with fusion. Which one would you pick for the product manual scenario above, and why?

There are two approaches:

1. **Unified embedding space:** Text and images are converted into embeddings that exist in the same vector space. A text query can directly retrieve either relevant text or images.
2. **Separate indexes with fusion:** Text and images are stored in separate indexes. The query is searched against both, and the results are combined before being sent to the LLM.

My choice: Separate indexes with fusion.

For the product manual use case, I would prefer separate indexes with fusion because it can help reduce computational cost. The images and text can be searched and handled separately, and their results can be combined when required. This approach also provides more flexibility.